CHAPTER 6

Data Collection Modes and Associated Errors

The *mode of data collection* refers to the medium that is used in a survey for contacting sample members and obtaining their responses to survey questions. Today, a number of modes are in use that can be classified in terms of three dimensions: degree of contact with the respondent, degree of data collector–interviewer involvement, and degree of computer assistance. A presentation of the various modes of data collection using this classification scheme is shown in Table 6.1. Note the number of computer-assisted interviewing (CAI) modes in the table. Use of the computer for collecting survey data continues to increase as new technologies are developed for communicating and interfacing with respondents in their homes, at work, and during travel (Nicholls et al., 1997). Some of these are discussed in detail in this section.

There are essentially three principal modes of data collection: face-to-face (or personal visit) surveys, where the interviewer and respondent are physically present during the interview; telephone surveys, where the interview is conducted by an interviewer over the telephone; and mail surveys, where the questionnaire is mailed to the sample members and returned by mail. Face-to-face and telephone surveys are referred to as interviewer-administered modes, whereas mail surveys are self-administered. Other modes of data collection we discuss include the use of administrative registers and direct observation.

In this chapter these modes are described in more detail and their main features are examined. There is a choice when it comes to data collection mode. With each mode there are relative advantages and disadvantages. Mode selection is often complex and related to the goals of the survey, mode characteristics, various design issues, and the methodological and financial resources available. In this chapter we present these advantages and disadvantages for a number of modes and describe how the choice of mode affects data quality.

Table 6.1 Data Collection Modes as a Function of Data Collector Involvement, Respondent Contact, and Degree of Computer Assistance[a]

	High Data Collector Involvement		Low Data Collector Involvement	
	Paper	Computer	Paper	Computer
Direct contact with respondent	Face-to-face (PAPI)	CAPI	Diary	CASI, ACASI
Indirect contact with respondent	Telephone (PAPI)	CATI	Mail, fax, e-mail	TDE, e-mail, Web, DBM, EMS, VRE
No contact with respondent	Direct observation	CADE	Administrative records	EDI

[a] ACASI, audio CASI; CADE, computer-assisted data entry; CAPI, computer-assisted personal interviewing; CASI, computer-assisted self-interviewing; CATI, computer-assisted telephone interviewing; DBM, disk by mail; EDI, electronic data interchange; EMS, electronic mail survey; PAPI, paper-and-pencil interviewing; T-ACASI, telephone ACASI; TDE, touch-tone data entry; VRE, voice recognition entry.

6.1 MODES OF DATA COLLECTION

In this section we review a number of different data collection modes listed in Table 6.1. Each mode is described and comments are made regarding what is generally known about its characteristics.

6.1.1 Face-to-Face Interviewing

Face-to-face interviewing is the oldest mode of interview since it does not rely on modern communication technologies. Because it provides for the maximum degree of communication and interaction between the interviewer and respondent, face-to-face interviewing is often associated with good data quality and is viewed by many survey researchers as the preferred mode of data collection for most survey topics. This view has been challenged in recent decades mostly because of the measurement errors associated with sensitive topics. Indeed, the list of topics that can be adversely influenced by the interviewer's presence is increasing as further research on interviewer effects is conducted. Let us consider some of the advantages and disadvantages of the mode.

Advantages and Disadvantages

Face-to-face interviewing is usually associated with high budget/cost surveys since it requires the interviewer to visit or meet with the respondent in a home, a workplace, or a public place. Travel is usually a high-cost component of this mode. In some cases interviewers may have to travel long distances, completing only one interview in a day's time. In general, face-to-face interview is usually more costly than other modes of data collection.

Face-to-face interviews are known to generate social desirability bias for some types of questions. As discussed in Chapter 4, social desirability bias is a phenomenon that may occur with sensitive questions. For some topics, there may be a tendency in face-to-face surveys for the respondents to be more concerned about how they are viewed by the interviewer than in providing accurate answers. This phenomenon manifests itself in various ways. Survey topics involving socially stigmatized behaviors such as consumption of alcohol or drug use or the number of sexual partners are typically underreported in face-to-face interviews. For attitude questions, the respondent might choose not to disclose extreme attitudes regarding political preferences or racial issues in the interview. There is also evidence in the literature that respondents tend to want to be perceived as being knowledgeable and up-to-date regarding current events in society when in fact that is not the case.

However, social desirability bias is not the only source of measurement error of concern in interviewer-assisted surveys. As noted in Chapter 5, interviewers themselves are an important error source. Interviewers have a tendency to affect respondents in ways that differ among interviewers. Each interviewer has his or her own set of behaviors, work procedures, question-delivery and question-wording techniques, probing techniques, strategies about when to probe, pace, and methods for recording responses to open-ended questions. Because of these differing interviewing styles, the interviewers as a collective contribute correlated interviewer error to the total survey error. As noted in Chapter 5, this error component can increase the standard errors of the estimates considerably and is not normally reflected in the variance estimates calculated in sample surveys.

One approach for minimizing interviewer effects is that of standardized interviewing procedures. The idea is to train interviewers so that they all perform in nearly the same way and therefore eliminate the potential for interviewer variance. However, as mentioned in Chapter 5, there is a controversy regarding the standardized versus conversational and other styles of interviewing. These issues suggest that the use of interviewers in surveys and methods for reducing their effects are still being researched actively in the field of survey methodology.

Despite these concerns about interviewer variance and social desirability bias, face-to-face interviewing is still considered the most flexible mode available, for a number of reasons. The mode allows for long, complex interviews that may last for an hour or more. Since the interviewer is in the presence of the respondent, he or she can use many tactics to gain cooperation, can make direct observations during the interview that can be recorded for later analysis, can ensure that the respondent's answers are not affected by the presence of others, and can provide probes for more complete and accurate answers when necessary.

As mentioned, face-to-face interviewers can administer visual aids and response cards for sensitive questions and can build rapport and confidence in their interaction with the respondent. Perhaps the most valuable feature is the fact that response rates can usually be kept at relatively high levels. The potential for face-to-face interviewers to gain cooperation, both initially and, if necessary, through refusal conversion is unmatched by modes with no face-to-face contact.

Another advantage of face-to-face interviewing is its high coverage of the general population for sample designs, such as area probability sampling. Area sampling entails sampling from an area frame such as a map or a photograph, and face-to-face interviewing is usually suitable for collecting data on those units that are sampled in the ultimate step (see Chapters 3 and 9). Despite the possibility for interviewers to control the presence of others during the interview, contamination by others might be a problem, both in the respondent's home and in other places, such as at work or in a public place. The presence of others can impede the respondent's ability to be completely open and honest in answering certain types of questions.

Perhaps the most serious practical problem with face-to-face interviews is the cost associated with its use. Since interviewers must travel to the units, workloads are much smaller than in telephone interviews. Thus, it usually requires more time and personnel resources per completed case than do, say, telephone surveys. In addition, the administration of training activities and observations of interviewer performance is time-consuming and costly.

Finally, an important concern associated with face-to-face interviewing is that the interviewer might falsify the interview. This behavior is referred to in the field as fabrication, cheating, curbstoning or table-topping (see Chapter 5). This behavior is known to all survey organizations and can be very difficult to detect. The interviewer may falsify an entire interview or simply skip questions during an interview to save time and respondent burden and then later fill in the skipped questions by making up data. A study of this phenomenon is found in Biemer and Stokes (1989). The behavior can be discovered by means of reinterviews or verification interviews and analyses of response distributions (see Chapter 8). One reason for interviewers to falsify interviews might be the difficulty in accessing certain population segments: for instance, high-crime areas or in apartment buildings protected by security staff.

Computer-Assisted Personal Interviewing

With the advent of systems for computer-assisted interviewing (CAI), the computerized variant of face-to-face interviewing termed CAPI (computer-assisted personal interviewing) is being used more and more. The interview is conducted by means of a program stored in a laptop computer. The program is such that survey questions appear on the screen in the order prescribed for each respondent. The interviewer asks the questions and enters the answers provided by the respondent.

Computer-assisted interviewing was first used in centralized telephone interview settings, referred to as computer-assisted telephone interviewing (CATI); see Section 6.1.2. In centralized CATI facilities, interviewers are housed together in large rooms, referred to as *telephone facilities* or *calling centers*, with up to 100 interviewing stations or more. Cases are typically assigned to interviewers from a centralized database on demand. When the interviewer completes a call, the call outcome is entered into the database. If the complete interview is not conducted, the case is made available for assignment to another interviewer to complete at another time. Usually, some fraction (e.g., 5 to 10%) of all interviews are monitored by a supervisor for quality control purposes.

In theory it would seem that using a computer program to control the interview would eliminate some errors. For example, if the questionnaire contains skips, the computer program can choose the correct route based on previous answers. Also, if the questionnaire for a household survey contains questions concerning more than one household member, the name of the person being asked about in the question can be filled in by the computer so that this information automatically becomes part of the question. Thus, the computer software automatically "personalizes" the questionnaire using "fills," which replace the standard generic language common in paper questionnaires with wording adapted for the particular respondent being interviewed. Questions concerning all vehicles owned by a household, events that are being investigated in some detail, and questions concerning all workplaces of a business can also be handled this way, thereby avoiding confusion on the part of both interviewers and respondents.

Certain data processing activities can also take place during the interview, such as online editing and coding (see Chapter 7). As discussed by Groves and Tortora (1998), these theoretical and logical advantages associated with CAI are not always supported by data from methodological studies in the sense that differences are always statistically significant. Nevertheless the studies usually show clear reductions in indicators of measurement error.

Effects on Data Quality

The literature on CAPI's effects on data quality is very sparse. There are some studies comparing face-to-face paper-and-pencil interviewing (PAPI) with CAPI regarding such process variables as skip error rates, item nonresponse, out-of-range responses, and interviewer variance, but the differences reported are not substantial. Instead, the evidence suggests that survey organizations and survey sponsors opt for CAPI for reasons other than data quality. Savings in costs and time are important, and such gains have been accomplished by some organizations. Other organizations, however, report that CAPI actually costs more than PAPI, but comparing the two modes on the basis of costs alone may not be relevant since CAPI is much more feature-laden than is PAPI. For example, CAPI provides greater opportunities to conduct interviews of much greater complexity than PAPI, to use dependent interviewing (where

responses obtained on earlier data collections can be part of the question wording), and to use advanced interviewer–respondent interaction in, for instance, online editing. Therefore, even if CAPI costs somewhat more than PAPI, the value added to the interview as a result of these features may be well worth the increased costs.

Converting from a PAPI to a CAPI Environment

For survey organizations, there are costs associated with the transition from a noncomputerized environment to a computerized one. These are startup costs for hardware, programming, and interviewer training. In some cases, CAPI programming errors are discovered after the interviewing has begun. It seems that many organizations have continuing problems with debugging CAPI applications and may occasionally experience a loss of cases due to errors.

Thus, there are a number of reasons to make the transfer from PAPI to CAPI, but CAPI cannot be used for all surveys and survey organizations. Sometimes the startup cost can be high even for a large organization, and moving from one CAPI system to another can be extremely burdensome. For organizations that are small, even commercially available CAPI systems might not yet be economically feasible.

For organizations that already have a CAPI system in place, it might not always be worthwhile to use the system for one-time surveys. Neither may it be feasible if the survey has to be carried out very quickly. There might not be enough time to do the technical work and train the interviewers. So most large organizations must be prepared to conduct both PAPI and CAPI, depending on the circumstances.

Face-to-face interviewing is expensive but highly flexible, provides high response rates, but may generate social desirability bias and interviewer variance.

6.1.2 Telephone Interviewing

Telephone interviewing has not always been accepted as a data collection methodology for social and economic research. In the 1936 U.S. presidential election, the results from a telephone survey based on telephone directory listings predicted a landslide victory for Landon over Roosevelt, which was published widely. When Roosevelt won, the telephone methodology was faulted for the erroneous prediction (Katz and Cantril, 1937). Actually, at that time, only 35% of the households in the United States had telephones, and the telephone population was disproportionately Republican, which explains the preference of telephone households for Landon.

Unfortunately, it took 40 years for telephone surveys to begin to receive acceptance among U.S. social scientists as a legitimate mode of interview. Two reasons for the increased interest in the mode are the lower cost of data collection than for face-to-face surveys and the increased coverage of the population by telephone both in the United States and worldwide. For example, Groves and Kahn (1979) provide a detailed examination of the cost and error components for an RDD and area frame face-to-face survey. They show that RDD telephone surveys can provide data quality which is comparable to face-to-face, but at a much lower cost. However, today, use of telephone surveys is again being challenged as a sole mode of data collection. The primary reason for this is the trend toward lower and lower response rates in surveys. However, the telephone is still used for many surveys, particularly in combination with other modes, called *mixed-mode surveys*.

Below, we discuss some of the advantages and disadvantages of this mode of data collection as a stand-alone mode as well as in combination with other modes of data collection.

Advantages and Disadvantages

Since they both use interviewers for communicating with the respondent, it is not surprising to find that the characteristics of face-to-face and telephone interviewing are very similar. Both modes have the potential to create interviewer variance and social desirability bias in the survey results. However, the literature suggests that these effects are somewhat less in the telephone mode (see Chapters 4 and 5).

Telephone interviewing is somewhat less flexible than face-to-face interviewing in that many things that are possible in face-to-face interviewing simply cannot be done over the telephone. For instance, visual aids cannot be used unless they are mailed to the respondent before the interview, which is a very impractical procedure. Also, questions delivered over the telephone cannot be too complicated and should not contain more than six but preferably fewer response alternatives.

Complexities such as asking the respondent to go through records and make calculations are extremely difficult to perform in a telephone interview setting. The social desirability bias might be smaller than in face-to-face interviewing because of the anonymity of the interviewer, but this very anonymity is not advantageous for developing rapport and persuading the respondent to participate or perform his or her best during the interview. As a result, there is a tendency for more acquiescence and extreme responses (choosing the endpoints of scales or choosing the first or last of response categories provided) over the telephone and less considered responses in general.

For example, the number of "don't know" and "no answer" responses tend to be larger over the telephone than in face-to-face surveys, and the faster pace leads to shorter replies. Also, interviewer variance might be larger than in face-to-face interviewing, due to the fact that telephone interviewers are usually assigned larger workloads than what is feasible in face-to-face interviewing. Typically, response rates are lower in telephone surveys than in face-to-face surveys of comparable type and size.

Telephone surveys tend to be considerably shorter than face-to-face interviews. Although there are examples of successful telephone surveys having average interview durations of one hour, these are quite rare for surveys of the general population. Most telephone interview designers try to limit the interview time to 30 minutes or less. Telephone surveys can be set up more quickly and are less expensive than face-to-face surveys.

Centralized telephone interviewing offers special advantages, since general administration, supervision, feedback, and training are more easily accomplished in such a setting. In some countries use of the telephone is not possible as the main data collection mode because of low rates of telephone coverage (referred to as *penetration rates*). The major problems associated with the telephone mode are that not all sample members have telephones, that answering machines and caller-ID equipment can prevent access to the sample unit, and that the questionnaire cannot be too long or utilize features such as visual aids and many response alternatives.

Computer-Assisted Telephone Interviewing

Today, telephone interviews are often conducted using CAI technology. In fact, CATI was the first computer-assisted data collection methodology to enter the scene about three decades ago and has now become standard practice in many organizations. It no longer belongs in the category of new technology. The benefits and drawbacks of CATI are very similar to those of CAPI, discussed in Section 6.1.1. There is also generally wide-spread agreement that both CATI and CAPI do not add appreciably to measurement errors simply by virtue of the computerization of the instrument (Couper et al., 1998). The key advantage of CATI seems to be the ability to avoid erroneous skips that plague complex paper questionnaires, and thus item nonresponse rates for CATI are generally lower than for PAPI.

TELEPHONE INTERVIEWING COMPARED TO FACE-TO-FACE INTERVIEWING

Advantages	Disadvantages
• Less costly	• Less flexible than face-to-face surveys
• Interviewer variance is usually smaller than for face-to-face surveys	• No ability to use a visual medium of communication
• Social desirability bias usually smaller than in face-to-face surveys	• Interviews must be shorter than in face-to-face surveys
	• No coverage of nontelephone units

6.1.3 Mail Surveys

In a mail survey a questionnaire is mailed to each sample member who is asked to complete the questionnaire and then send it back to the survey organization. Since no interaction with a data collector is involved, the mode is referred to as *self-administered*. As such, it is imperative that the questionnaire be self-explanatory to the respondent (i.e., questions and instructions must be easily understood by the respondent since there is no help available from the outside). To a much greater extent than for interviewer-assisted modes, the quality of the resulting data hinges on the quality of the questionnaire design.

In addition, as discussed in detail in Chapter 3, the response rate to mail surveys can vary tremendously by survey organization, owing, to a great extent, to the skill and knowledge of the survey staff. The care that is needed to implement mail surveys successfully is quite evident from the discussion in that chapter. In this section we describe some of the advantages and disadvantages of mail surveys compared to other modes of data collection.

Advantages and Disadvantages

During the last 20 years the mail survey has gone through a revival. As the cost of the interviewer-assisted modes has increased considerably, mail surveys have become an attractive option. Mail surveys are quite inexpensive to implement, which makes them the preferred mode for low-budget surveys. In addition, mail surveys may have some quality advantages over interviewer-assisted modes, due to the reduced risk of social desirability bias associated with self-administration. Since the visual communication medium is available for this mode, it has advantages over telephone surveys for situations where providing the respondent with a visual aid is necessary: for example, the use of maps to indicate various sites visited on a recent trip to a national park or for questions that contain more than, say, six response alternatives.

Since the respondent has possession of the questionnaire, mail surveys also provide time for the respondent to give thoughtful answers and look up records. Finally, mail surveys do not generate the kind of variance contribution from interviewers, and errors generated by question order or response alternative order are reduced (Dillman, 2000).

However, mail surveys are known to generate lower response rates than do interviewer-assisted modes. Therefore, it is usually combined with follow-ups conducted by interviewers (CATI or CAPI mixed-mode surveys). A few decades ago the mail survey was considered inferior to interviews because of the low response rates and the fact that the survey organization has little or no control over the response process. For example, it is not possible to ensure that the intended sample person completes the questionnaire or that he or she collaborates with others as appropriate and as intended in the survey design. When the questionnaire is not returned, it is not known whether this is because the questionnaire never reached the sample person or if it did reach the sample person but he or she failed to return the questionnaire. In some cases, the sample person may not even be eligible to complete the questionnaire, yet no information is available for the survey staff to make that determination. This is problematic for both the computation of response rates and adjusting the estimates for nonresponse. There is also a greater risk of considerable item nonresponse rates associated with the mode.

For household surveys, mail surveys present problems when the sample person is to be selected at random from within the household. Although schemes such as the last birthday method (Chapter 9) have been tried, there is concern that these methods do not produce random samples and should not be used in rigorous scientific research. Therefore, mail surveys are not recommended in situations where the questionnaire cannot be mailed directly to a specific, named respondent.

Mail surveys also require a long field period to obtain acceptable response rates. Usually, eight weeks or more is required from mailing the first mailing to the final return. Also, the questionnaire has to be simple in the sense that the respondent understands the concepts and can navigate through the questionnaire without problems. Of course, respondents must be on a literacy level that allows them to read the questions and the instructions. In the United States a general rule is used that the reading level of the questionnaire should be at a fifth-grade level for mail questionnaires used in government surveys of the general population.

Recent Research

A considerable part of the revival in the use of mail surveys can be attributed to the work of Don Dillman at Washington State University in Pullman, Washington, U.S.A. (see, e.g., Dillman, 1978, 2000). Dillman has developed efficient strategies for mail survey data collection, consisting of step-by-step operations that, if followed, will result in response rates that are larger than for other documented strategies. Dillman's strategies use social exchange theory as a basis and were described in Chapter 3. He has also started to develop a theory for self-administered modes based on theories of graphic language, cognition, and visual perception combined with the theories used by Cialdini (1984) and Groves and Couper (1998) to increase cooperation in interview surveys. Some of these social and psychological influences (reciprocation, commitment and consistency, social proof, authority, scarcity, and liking), discussed in Chapter 3, can also be used in the design and conduct of mail surveys.

Thus, there are two sets of principles at play. One concerns development of the questionnaire in such a way that it becomes self-contained, so that the organization of the information in the questionnaire and the navigational guide offered to the respondent are such that completion of the questionnaire is ensured. Viewed from this perspective, it appears that these features aid the respondent in much the same way as interviewers guide respondents in interviewer-assisted modes.

The second set of principles involves implementation of the data collection in a prescribed, step-by-step fashion. This involves the use of advance letters, multiple follow-ups, real stamps on the return envelopes, personalized mailings, progressively urgent appeals to the respondent, and other strategies for increasing response rates. These principles are described in Chapter 3 as well as in much more detail in Jenkins and Dillman (1997), Dillman (2000), and Redline and Dillman (2002). The principles have been and will continue to be revised as new information is gained through experimental studies.

A *mail survey* does not generate interviewer variance and is very suitable for collecting some types of sensitive information. Although inexpensive, a mail survey takes time to carry out, often produces relatively low response rates, and the risk for item nonresponse is considerable.

6.1.4 Diary Surveys

Diary surveys are used for the purpose of collecting information on events retrospectively. The structured questionnaire is replaced by a diary where respondents enter information about frequent events, such as household purchases, food intake, daily travel, television viewing, and use of time. To avoid recall errors, respondents are asked to record information soon after the events have occurred. Often, this means that respondents should record information on a daily basis or even more frequently. Thus, successful completion of the diary requires that respondents take a very active role in recording information.

Diary surveys differ from mail surveys because interviewers are usually required to contact a respondent at least twice. At the first contact, the interviewer delivers the diary, gains the respondent's cooperation, and explains the data recording procedures. At the second contact, the interviewer collects the diary, and if it is not completed, assists the respondent in its completion. Because of the need for a high level of commitment from the respondent to maintain the diary accurately, diary reporting periods are fairly short, typically varying in length from one day to two weeks.

For example, when the purpose of the survey is to collect data on household expenditures, the respondent is asked to record the specific purchase as soon as possible after its completion, together with the price of the merchandise. Such detailed information cannot be collected via interviews because the respondent will not remember some purchases, particularly small ones, for more than a few days. The interviewer is a critical component of the method since for some types of purchases, interviewers can assist the respondent in providing more accurate information than the respondent alone can be expected to provide.

Particularly for large purchases, such as cars or boats, the risk of *telescoping* effects is high (see Chapter 4). Such purchases are salient events to many respondents, and there is evidence that such events have a tendency to be telescoped forward in the reporting period. Some respondents may disregard the

payment of bills as purchases, and consequently, in some expenditure surveys, credit card payments are considered out of scope. Instead, the respondent is asked to record each single purchase at the time it is made.

Despite efforts to stimulate cooperation by interviewer contacts, incentives, reminders, and so on, there is a clear tendency for respondents gradually to lose interest in the recording task, as evidenced by fewer purchases reported at the end of the collection period than at the beginning (Silberstein and Scott, 1991).

There is also a clear risk of conditioning on the part of respondents. As a result of participation in the survey, the respondent might change his or her purchasing behavior. For instance, when the respondent notices the amounts spent on specific goods, he or she might change the purchase pattern temporarily. Soon after the data collection, the purchase pattern is usually back to normal, resulting in biased estimates.

For many items, recall errors are such that not all purchases are recorded, resulting in underestimates of purchases. It is very difficult for respondents to think constantly about keeping receipts from eating out or buying gas. Thus underreporting is a big problem in these areas. These recall errors increase with the length of the data collection period.

In some designs proxy reporting is a problem. If there is one diary respondent per household, underreporting will occur. Personal diaries, of course, eliminate the effects of proxy reporting. Social desirability bias is smaller in diary surveys than in interviews but can still be a problem, especially if there is just one diary per household. Typically, purchases of alcohol and tobacco are underreported in expenditure surveys. Also, as in all paper-and-pencil self-administered modes, the literacy and education levels are important prerequisites to obtaining good-quality data.

There is evidence that the design of the diary can be an important error source. There are essentially two designs. One consists of more-or-less blank pages, and the other is more structured, with preprinted headings or sections. Comparisons between the two have provided inconclusive results (Tucker, 1992). Recently, two issues for the design of diary surveys have been discussed extensively in the literature. One issue concerns the optimal length of the recording period. Years ago, four-week diaries were used for purchases in household expenditure surveys. However, in recent years, this practice has been abandoned because of the extensive respondent burden involved and the fact that the completeness of responses declines during such a long period. As a result, shorter collection periods for diary surveys are common today.

Another issue concerns the types of events that can be studied using diaries. A recent trend is to limit the scope of the survey to avoid the heavy respondent burden involved. There are examples of expenditure surveys that collect data on food items only. However, this trend is not present in all countries. For instance, one design used in Holland is such that it consists of two diary components. The first is a two-week complete diary and the second is a one-year recordkeeping of expenses above a certain amount.

A specific feature associated with diary surveys is that the placement of diaries follows a prescribed pattern. Typically, the sample is dispersed in time throughout a whole year to accommodate for the collection of all kinds of purchases or all kinds of activities. If surveys like these were restricted to parts of the year, the occurrence of some events would be heavily underestimated, due to seasonal patterns. Therefore, data collection is conducted by dividing the sample into subsamples that start their recording at prespecified dates. It is important that there is an even distribution of the sample, even by a prespecified day of the week. Practical reasons usually prevent exact adherence to the placement plan. Most organizations therefore allow minor deviations from the plan. One must bear in mind, however, that letting a sample member start the recording at a date other than the one prescribed by the design will result in measurement errors due to "nonresponse in time" even though actual nonresponse is avoided by such a decision (Lyberg, 1991).

6.1.5 Other Self-Administered Modes

There are a number of computerized versions of self-administered modes that have helped enhance the usefulness of self-administration (see Ramos et al., 1998). *Electronic data interchange* (EDI) is a technique to obtain economic and other data directly from sampled businesses' own computer-stored records. Although still in its infancy, the long-term goal is for the survey organizations to collect data from these records, thereby reducing the need for traditional surveys in some fields. Obviously, there are a number of problems to solve before widespread use of EDI can be anticipated. There is a need for convincing businesses to use standard messaging formats that can be used to transfer statistical data to the collecting agencies. Experience shows that standardization can be difficult to accomplish since not only businesses are involved but also developers of accounting software. But the very idea of linking a business's database to the collecting organization is an appealing one and shows great promise since businesses always are interested in less paperwork (Keller, 1994).

In *disk by mail* (DBM), a diskette is sent to the respondent via postal mail. The diskette contains a self-administered questionnaire and respondents install and fill out the questionnaire on their own computers. The responses are saved on disk and mailed back to the survey organization. DBM works almost like computerized interviewing since the DBM program controls the question flow, provides instructions, and performs edit checks. Clearly, the use of DBM has the potential of saving costs and reducing errors, and if viewed as an alternative to the interviewer-assisted mode, no interviewer variance is involved.

Electronic mail survey (EMS) is a mode where computerized self-administered questionnaires are transmitted to respondents and returned via electronic mail or delivered via the Internet or the World Wide Web (WWW). DBM and EMS have many elements in common. The basic difference is that

Web surveys have current or potential features that DBM and electronic mail do not have. With the graphic and multimedia capabilities of the WWW, the design choices are almost unlimited, and it is important for survey mode researchers to intensify their work on developing guidelines for Web survey design (see Couper et al., 2001). There are currently two lines of development. One is led by large survey organizations and is characterized by a careful step-by-step development that appreciates the fact that not everybody has access to or wants to use the computer to answer survey questions. Therefore, DBM and EMS are usually offered as alternatives to other collection modes by large survey organizations. Another line of development is led by organizations that might not even have survey work as a business. Examples of the latter are newspapers and TV networks conducting investigations called, for example, "Today's Web question," resulting in an abundance of "surveys" based on self-selection.

DBM and EMS have a number of limitations, which have been summarized by Ramos et al. (1998):

1. There is no readily available software for electronic questionnaires that suit survey organizations using capabilities found in most CAPI and CATI programs, such as skips, edit checks, and randomization of response alternatives. The software has to be developed in-house.
2. The questionnaire design capabilities are limited, which restricts the scope of applications.
3. E-mail is based on nonstandardized software, with no standardization in sight.
4. Respondent access to modems and communication software is limited.
5. E-mail addresses often change.
6. There are many issues associated with confidentiality and security. Confidentiality must be assured and guarantees must be provided that installment of the communication package will not lead to virus attacks.
7. Coverage error is a concern, due to limited respondent access to properly equipped PCs and limited modem and Internet access.

Despite the limitations, there are reasons to expect measurement error to be reduced by DBM and EMS compared to traditional mail surveys. These reasons are related to those behind CATI and CAPI improving traditional interviewing. DBM and EMS provide a more controlled collection environment. Controlled routing and embedded edits decrease measurement error and item nonresponse. There is also evidence that answers seem more thoughtful in DBM and EMS applications.

Touch-tone data entry (TDE) is a collection mode in which the respondent calls a computer and responds to questions asked by the computer. The answering process is such that the respondent enters data by using the keypad of a touch-tone telephone. After entering the response, the computer asks the respondent to verify the answers provided.

TDE is a very common communication technique in today's society. It is used for ordering, transactions, and routing of telephone calls. At this time, TDE has only been used in surveys that are quite short (say, 5 to 10 minutes) and data entry is simple. It is particularly suited for situations where the information requested is either numeric or can be transformed to a numeric code. TDE is an alternative to a mail questionnaire or an interview and the development has been triggered by concerns associated with costs. Also, TDE allows respondents to choose the time for interview since the service is open 24 hours a day, seven days a week.

Another advantage is that the collection time per respondent is shortened in many TDE applications compared to manual modes. For the collecting agency, it is advantageous to eliminate interviewer involvement, not only because of costs but also for practical reasons associated with the 24-hour access for respondents. The obvious disadvantage is the limited scope of application. Since TDE works best for data collection situations involving short, numeric, and repetitive entries, there may be few surveys where it can be used successfully.

Voice recognition entry (VRE) is a variant of TDE. Instead of using a keypad to enter a response, the respondent speaks the answers into the telephone and the computer verifies it by repeating the response. This technology has become fairly common for very simple exchanges of information in a number of situations in society. Some examples of VRE applications include banking by phone, automated order entry, automated call routing, and voice mail. VRE is more difficult to apply since spoken answers are more easily misinterpreted or not recognized by the computer. On the other hand, the technology works for any kind of telephone, and respondents seem to think it is simpler than TDE (see Clayton and Werking, 1998).

As summarized by Dillman (2000), TDE and VRE are limited technologies. The respondents lose the overview they have in mail questionnaires, and they do not get the help that is provided by interviewers in interview surveys. The design of TDE and VRE is complicated since all situations must be anticipated in the script, and appropriate messages must be formulated for these situations. On the positive side, all stimuli are delivered in a completely uniform way, which is not possible with "live" interviews.

Within the last decade, a new mode of data collection and its variants has gained popularity, particularly in surveys dealing with sensitive topics. This mode is *computer-assisted self-interviewing* (CASI). As discussed earlier in this chapter, interviewer-assisted modes are generally not recommended when the survey deals with sensitive topics and other topics prone to social desirability bias. CASI is a computer technology that combines the benefits of face-to-face interviewing, self-administration, and computer assistance. With CASI, part of the interview (e.g., questions that are not sensitive and could benefit from interviewer assistance) can still be conducted by the interviewer. However, when sensitive topics arise, the computer can literally be turned around to take over the interview and guide the respondent through the sensitive questions.

Recall that one early low-tech solution to this problem described in Chapters 3 and 4 is the *randomized response* (RR) *technique* (Warner, 1965). With the advent of computer-assisted collection methods, the RR method has been replaced by techniques in which a laptop computer is essentially turned over to respondents for part or all of an interview. In CASI (sometimes referred to as *text-only CASI*), respondents view the questions on a computer screen and enter their answers using a computer keypad. This form of CASI requires respondents to have a certain literacy level to be able to read the questions. This requirement is eliminated with *audio CASI* (or ACASI). Using ACASI, respondents can both listen to questions using headphones and read the questions on the screen and then enter their answers by pressing labeled keys. Both (text-only) CASI and ACASI require that the interviewer deliver the instrument on the CAPI laptop computer and then wait until the respondent has finished the task of answering the questions to retrieve the computer. A natural and cost-effective alternative is to implement computerized self-interviewing in a telephone setting. *Telephone audio CASI* (T-ACASI) is almost identical to TDE and can be either through respondent call-in or through interviewer call-out. The primary difference is that with T-ACASI, an interviewer initiates the call and then turns the interview over to the computer and the respondent for the remainder of the interview. TDE usually does not involve an interviewer. Calls are either made by the respondent to a computer or by the computer to a respondent.

The CASI variants originated from the need for good measurements of sensitive attributes. Viewed from that perspective, they seem to be successful since they generate increases in reporting (see Tourangeau and Smith, 1998; Turner et al., 1998). The CASI development differs fundamentally from CATI and CAPI. The latter technologies changed the interviewer role so that it became more standardized. CASI changes the interviewer–respondent interaction completely and the interviewer assumes a more administrative role, while the collection format is still an "interview" with all advantages associated with that mode compared to the mail format. As discussed by Turner et al. (1998) and Tourangeau and Smith (1998), audio-CASI seems to reduce underreporting of certain sexual behaviors as measured not only by face-to-face interviews but also by self-administered paper questionnaires.

6.1.6 Administrative Records

For some data collection objectives, the use of administrative records is a cost-effective and high-quality alternative to surveys. *Administrative records* are records that contain information used in making decisions or determinations or for taking actions affecting individual subjects of the records. Most nations keep records to be able to take actions regarding licensing, taxing, regulating, paying, and so on. It is, of course, very tempting to use such data for statistical purposes, and there are indeed a number of such register-based collections in countries with a tradition of record-keeping. In some countries the use of administrative records for statistical purposes has a long history. As mentioned in Chapter 1, early censuses had administrative purposes, and some vital record systems in Europe are more than 300 years old. Recent advances in computer technology and record linkage methodology have opened up new possibilities to use records not only for primary collection purposes but also as auxiliary information in some estimation processes.

Thus, the situation usually is as follows: In some countries authorities and organizations build administrative records about a group of persons, businesses, or other entities. The purpose is to take action, if necessary, regarding individual entities. In a taxation register, all individuals and businesses are recorded and each record contains information regarding a number of variables, such as occupation, income, age, type of business, number of employees, address, number of work sites, total revenue, and so on. The organization in charge of the register needs information about the subjects so that the correct action can be taken regarding each subject, such as calculating the correct amount of tax for each person or business. Obviously, with a record like this it is possible to move from an individual-related action situation to one where one is interested in characteristics and attributes of groups of individual entities (i.e., statistics based on register or record data). For instance, it would be possible to provide tables on occupation distribution or via record linkage match survey data with occupation register data.

In principle, there is no difference between error structures found in administrative data compared to data collected via other modes. There is a tendency toward an unjustified reliance on the quality of administrative data among producers of statistics. In some applications concerning the evaluation of survey data, administrative record data are used as the gold standard. Sometimes this is justified since the administrative data have been collected under completely different circumstances than the corresponding survey data. But on other occasions, the data quality of the administrative record might be far from a gold standard.

The major concerns with data quality in administrative records include the following:

- In most cases there has been no influence on administrative data generation, updating, and coding on the part of the survey researcher interested in using the administrative data. For example, occupation descriptions provided by taxpayers are usually taken for granted by the record keeper since that variable is of marginal interest to the record keeper or is used just to screen certain segments of the population for greater scrutiny without the ambition to cover the segment entirely.
- Those in charge of administrative records usually do not have a statistical background, resulting in processes that are, at least for some variables, imperfectly collected, recorded, and controlled.

- Usually, there are conceptual differences between the statistical application and the administrative use.
- The quality of the administrative data is seldom assessed.

The major advantages associated with the use of administrative data are:

- The records and the data already exist and are inexpensive to use.
- In many applications there is no need for contacts with register units and therefore there is no respondent burden.

The use of administrative data is discussed further in Zanutto and Zaslavsky (2002) and Hidiroglou et al. (1995b).

6.1.6 Direct Observation

Direct observation as a mode of data collection is the recording of measurements and events using an observer's senses (vision, hearing, taste) or physical measurement devices. In this mode the observer assumes the role of the respondent. Direct observation is not uncommon in survey research. Applications include estimates of crop yield, counting number of drivers not using their seat belts, measuring TV viewing by electronic devices, measuring pollution via mechanical instruments, and assessing land use by interpreting aerial photos. Direct observation also has a long tradition in anthropology and psychology. Some of that tradition has been transferred to recent cognitive survey research where respondent and interviewer behavior is studied via observations.

The error structures in direct observation are of two kinds. When observers use their senses, the error pattern that emerges is very similar to that generated by interviewers. Observers may misperceive the information to be recorded, they exhibit variability within and between themselves, and their behavior may change over time. The same types of control programs used for interviewers can be used for observers. The second kind of error structure emerges when mechanical devices are used for measurement. Studies using these devices must have a program for validation and instrument calibration to ensure that systematic errors do not occur (see Fecso, 1991).

6.2 DECISION REGARDING MODE

In every survey, a decision must be made as to which mode of data collection should be used. A number of factors enter into this decision process, including the desired level of data quality, the survey budget, the content of the survey instrument (question type, number of response alternatives, number of questions, complexity, and the need for visual aids), the time available for data collection, the type of population that is being studied, and the information available on the sampling frame for this population. It is also necessary to view the data collection together with other steps in the survey process, such as access to the sample units for data collection and data processing alternatives available.

Often, however, the mode decision is clear because many alternatives are deemed unrealistic from a financial point of view or are otherwise not practical for the particular study. Examples of situations with only one mode alternative are the collection of data on expenditures of different kinds and the collection of data on crop yields. In the former case the literature uniformly suggests the use of the diary method because to keep track of purchases, it is necessary for the respondent to make notes of purchases in a continuing fashion to avoid major problems with recall effects. With the diary method it is possible for respondents to record purchases soon after they have been made. Any other known data collection method is unsuitable because respondents cannot possibly recall all purchases made during a specific time period unless they concern goods such as houses, cars, washing machines, and other investmentlike purchases. In the latter case with crop yields, the only possible collection methods are eye estimates made by observers or harvesting crops on sampled plots.

The mode decision is more complex when several alternatives exist. Some decision factors are relatively easy to assess. For instance, the cost can usually be predicted relatively easily, and so can the nonresponse rates given that a set of relevant design factors has been established, such as sampling method and the system for nonresponse follow-up. When measurement errors are crucial to the designer, the decision becomes more difficult.

There are two ways of deciding which data collection mode is best in terms of minimizing the measurement error (for reviews and additional references, see Lyberg and Kasprzyk, 1991; de Leeuw and Collins, 1997). One method involves the application of a set of general principles or "rules of thumb" associated with each mode. For example, if strict sample-size requirements are placed on the survey, it might be difficult to use a mail or a face-to-face survey, since they cannot be adjusted easily for over- or underestimates of response rates. When the survey budget is low, a mail survey might be the only option. Complicated questionnaires with branches and skips do not favor mail surveys. If social desirability bias is a concern, interviewer-assisted modes could be problematic. Questions with many response alternatives do not function well in an interview setting, but in face-to-face interviewing there is the ability to use flashcards (i.e., cards that show the response alternatives to the respondent, thereby creating a "self-administered situation"). If a high response rate is crucial, a face-to-face interview is to be preferred. If a high-degree of geographic dispersion is necessary (e.g., samples with very little geographic clustering of the sample units), face-to-face interviews are usually too expensive, due to travel costs. Interviewer control is easier to carry out in a centralized telephone interview setting than in a decentralized telephone interviewer or a face-to-face interviewer organization. The list of general rules of thumb goes on and on.

Typically, the decision is made on the basis of the priorities of the researcher as to how he or she values the various advantages and disadvantages of each option. Prior research on mode comparisons are usually quite helpful in making these choices. If there is very little information in the literature regarding which mode of data collection is preferred for the particular set of phenomena to be studied, it may be necessary to conduct a study to address that issue.

In designing a mode comparison study, there is a choice as to what type of mode effect should be measured: a pure mode effect or a mode system effect. A *pure mode effect* is essentially a measurement bias that is specifically attributable to the mode. To measure the pure mode effect accurately, one would have to compare the results of two collections where the only difference between the collections is the mode (i.e., the same questions are asked in the two collections, the same characteristics are measured, the same population is studied during a specific reference period, etc.). The resulting difference between the response distributions generated by the two collections would then constitute an estimate of the mode difference (Biemer, 1988; Groves, 1989).

However, in practice things are more complex. An estimate of the kind described must rely on a laboratory experiment or a field test whose realism is doubtful (see Chapter 10). Very rarely do we have a situation where it is possible, feasible, or desirable to keep all design factors intact and just vary the mode. This may be possible only in very simple and unrealistic applications, which may have little relationship with the complex realities of large-scale data collection. Further, the survey practitioner's main interest usually lies not just with the pure mode effect, but also with the combined effects of mode and other design factors. The following example illustrates the difficulty and undesirability of experimenting pure mode effects in practical situations.

Example 6.2.1 Suppose that we wish to compare telephone and face-to-face interviewing. Interviewer skills are probably different in a face-to-face interviewing organization than in a more-or-less closely monitored centralized telephone interviewing organization. The sample population for a telephone survey might differ from that for a face-to-face survey, simply because not all persons can be contacted via the telephone, resulting in coverage differences. A face-to-face questionnaire that uses visual aids shown by the interviewer cannot easily be transformed into one that can be used in a telephone interview. Thus, if a survey practitioner considers converting from, say, a face-to-face interview to a telephone interview, the mode is not the only thing that changes in the conversions. To conduct a fair test of telephone interviewing, design factors such as interviewers, questionnaire, the field period, performance monitoring and supervision, and so on, should be optimized for the telephone mode. This suggests that many other design factors other than the mode of interview should vary between the two modes.

One way out of this dilemma associated with mode comparisons is, instead to measure *a mode system effect*: to focus on comparing whole systems of data collection rather than trying to isolate the effect of the mode. A mode system effect is easier and more realistic to estimate than a pure mode effect. By *system* we mean an entire data collection process designed around a specific mode. A system might include design factors such as interviewer hiring, interviewer training, interviewer supervision, questionnaire contents, number of callback attempts, refusal conversion strategies, sampling system, and sampling frame coverage. The difference between two systems can be measured along a number of dimensions, such as the response distribution, the unit nonresponse rate, the item nonresponse rate, the time needed to accomplish a given response rate, and the cost of the data collection. Some of these dimensions provide obvious comparative measures since a high response rate is better than a low response rate, a low cost is better than a high cost, and a short production time is better than a longer one. When it comes to the distribution of responses for certain items (i.e., the dimension that reflects measurement bias), we cannot assess a system difference unless we have some gold standard with which to compare the two outcomes. For some characteristics it is possible to declare one response distribution outcome as superior to the other. For example, sometimes we assume that more frequent reporting of certain behaviors or events is a sign of less recall error or less social desirability bias. In those cases "more is considered better," but one can never be completely sure unless an evaluation study is performed (Moore, 1988; Silberstein and Scott, 1991).

In general, data collection systems do not consist of one mode only, since mixed-mode surveys are the norm these days. For example, a common situation is to start with a relatively inexpensive mode, such as mail, and after a number of follow-up attempts, use telephone interviewing for the mail nonrespondents. In some surveys, face-to-face interviews have also been introduced to follow up mail nonrespondents that do not have telephones. There are also examples of surveys that begin with an interviewer-assisted mode and then a mail survey for follow-up for the mail nonrespondents. Some respondents prefer being sent a mail questionnaire rather than being interviewed (Carrol et al., 1986).

Even though the issue of a pure mode effect might seem of academic interest only, it quickly becomes important when we consider the situation above. Thus the fact that most surveys are forced to utilize more than one mode directly from the start makes the mode effect an important design consideration. The first step is to choose a *main* mode that can best accommodate the survey situation, where all important design factors are taken into account. The main mode is used to its maximum potential with certain constraints, such as a prespecified number of callbacks or reminders; then another mode is used

for the principal purpose of increasing the response rate. This is a necessary and most practical approach but is a strategy of compromise. If a particular main mode is chosen because it is deemed appropriate for the survey measurement situation at hand, a switch to another mode should result in data of lesser quality. It is therefore important to make sure that the size of the effect of using complementary modes is as small as possible.

In some surveys the mode effects are small because the same questionnaire can be used across all modes. Most problems occur when mail is combined with an interviewer-assisted mode. Unless the mail questionnaire is so simple that it can readily be used in interviewing, we are in trouble. A mail questionnaire with many response alternatives is not suitable in a telephone interview. If there are, say, eight response alternatives for a specific question, it is very difficult for the respondent in a telephone follow-up to remember them all and distinguish between them. It is also difficult and sometimes awkward for the interviewer to read all response alternatives. In these cases there is a risk that the respondent uses a response strategy that favors the first few response alternatives or that the respondent interrupts the interviewer as soon as a reasonably fitting alternative has been read by the interviewer and that the interviewer skips reading the remaining alternatives to achieve a smooth interview.

If a mail questionnaire has been chosen because some questions are very sensitive, any conversion to an interview-assisted mode may be risky. The conversion could result in larger measurement errors, and that loss in response quality must be weighed against any gains that may result in response rates. When a specific mode has been chosen for its measurement characteristics, a switch to another mode to, say, increase response rates can be very difficult to address in the survey planning stage; often, ad hoc strategies with minimal control are adopted to handle this trade-off situation. It is easier to plan for questionnaire format issues. Questions in a mail questionnaire can sometimes be constructed so that they fit both interview and self-administered situations. Unfortunately, this often means that the measurement potential is lowered for all modes involved (see Chapter 4).

A conversion from interviewing by telephone to face-to-face interviewing is usually less problematic. A face-to-face interviewer usually has more options to convert refusals and more opportunities to guide and assist respondents than does a telephone interviewer. Also, a face-to-face interviewer can provide visual aids, something that is usually not possible with telephone interviewing.

We have discussed the situation with a mixed-mode strategy where one mode is combined with one or two other modes to achieve a higher response rate. There are also other instances where mixed-mode strategies are appropriate. One application occurs when one mode is used to screen a population for a subpopulation with some rare attribute that we are interested in, and a second mode is used to measure that attribute and related ones for the subpopulation. For example, a telephone survey may be used to identify businesses in the United States that export goods to Sweden. The resulting subpopulation is then surveyed by means of a mail questionnaire or a face-to-face interview.

A third application of a mixed-mode strategy is rather common in panel surveys, where the purpose of the strategy is to reduce costs. Panel studies are surveys that return to the same sample units at periodic intervals to collect new measurements to measure changes over time at the respondent level. Each new round of interviewing constitutes a "wave." In the Current Population Survey conducted in the United States, a mixture of face-to-face and telephone interviews is used (U.S. Department of Labor and U.S. Bureau of the Census, 2000). Mixed-mode strategies in panel surveys try to take advantage of the best features of each mode. Typically, face-to-face interviewing is used in the first wave of the panel survey, which improves on the otherwise incomplete coverage associated with nontelephone units while establishing a relationship with the respondent to maintain cooperation and response in future rounds of the panel (Kalton et al., 1989). Telephone interviewing (possibly decentralized, that is, telephone interviewing from interviewers' homes) is used in subsequent waves unless the respondent prefers the face-to-face mode or does not have access to a telephone.

During the last couple of decades, data collection methods in surveys have undergone a technological transfer. To a large extent, data are now collected by computer-assisted means. The telephone is still a dominating mode, but self-administered modes have seen a revival for several reasons. First, it has become apparent that self-administration is the best mode when it comes to collecting certain types of sensitive data, because social desirability bias is small. Second, it turns out that self-administered modes can utilize computer assistance, and third, compared to interviews, self-administration is inexpensive.

In many countries, however, the literacy level is so low that self-administered modes are impossible to use. Also, telephone coverage rates are still very low in many countries, including developing countries, where the survey researcher must rely on face-to-face interviewing when surveying human populations. In some environments there is only one option of data collection mode, face-to-face interview, no matter what the survey topic is. Thus the survey world looks very different depending on the country and the population to be surveyed. In Table 6.2 we have provided examples of factors that dictate the choice of main mode.

6.3 SOME EXAMPLES OF MODE EFFECTS

We have seen that a study aiming at estimating the pure mode effect is both difficult to design and not necessarily relevant. Instead, mode comparison studies are usually designed to compare whole systems of data collection (i.e., the outcomes of each system with all its components are assessed regarding a set of relevant evaluation criteria). Examples of such criteria are unit and item nonresponse rates, completeness of reporting, similarity of response

Table 6.2 Factors to Consider When Determining the Optimal Main Data Collection Mode

Factor	Implication for Mode Choice
Concepts to be measured	If a visual medium is required, a telephone survey can be ruled out. Complex concepts usually benefit from interviewer assistance.
Target population to be surveyed	Can the nontelephone population be ignored? If so, consider the telephone mode. Literacy level: Mail modes require literacy rates at or above the national average.
Contact information available on frame	If a name and address, mail or face-to-face interview should be considered. If a telephone number, consider a telephone survey.
Saliency of the topic	If much persuasion is needed to obtain adequate response rates, mail surveys must be ruled out.
Speed of completion	If needed very quickly, telephone is best. If needed in weeks, a mail survey may be feasible.
Scope and size of the sample	If a local survey, all three main modes may be feasible. For a national survey, cost may be the reigning factor that suggests mail or telephone survey.
Sample dispersion	Maximum dispersion suggests a mail or telephone survey. In face-to-face surveys, some clustering is almost always needed.
Frame coverage of target population	If only poor coverage frames are available, use a face-to-face survey, RDD, or mixed-mode. Good coverage implies flexibility regarding mode choice. Availability of an address permits advance letters and prepaid incentives.
Nonresponse	Interview modes usually generate higher response rates than self-administered. Ability to persuade reluctant sample units depends on richness of media (e.g., in mail surveys, motivation is limited to written materials). Nonresponse is confounded with coverage problems in mail and telephone modes. Mail questionnaires might be regarded as junk mail and thrown away by sample unit. Mail questionnaires cannot be completed by parts of human populations, due to literacy problems.
Interviewer	Interviewer can generate response errors, such as social desirability bias. Interviewer-assisted mode is not good for collecting sensitive information. Interviewer necessary for visual aids and probing. Centralized telephone interviewing reduces costs and errors compared to noncentralized interviewing. Telephone interviewers can have larger workloads due to no travel burden.
Respondent	There is some evidence that respondents prefer self-administered surveys. Self-administered modes are suitable for collecting sensitive information. If the response task is difficult, interviewer assistance is necessary.
Instrument	Mail questionnaires must be relatively simple but are suitable when questions contain many response alternatives. Complex instruments call for the interview mode. Mixed-mode must use questionnaires that can be used in all modes.
Cost	Everything else may be secondary if mail is the only mode that can be afforded. The telephone interview mode is less expensive than the face-to-face mode.

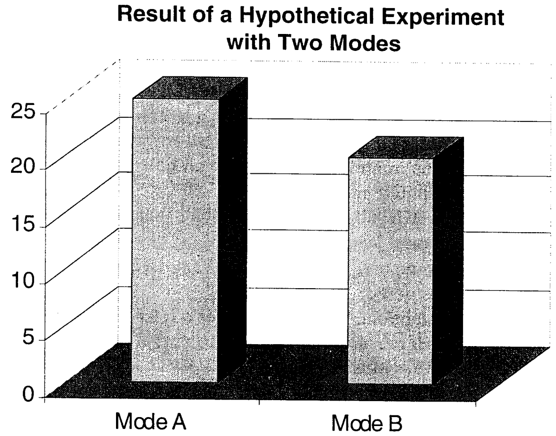

Figure 6.1 Which mode is better? The results of a mode comparison study may be inconclusive. In this hypothetical example, the two modes yield significantly different estimates of the parameter, but this information alone is inadequate to determine the better mode (i.e., the mode that yields an estimate closer to the true parameter value). However, if information is available on the direction of the bias for the characteristic under study, the better mode can be identified. For example, if we know that the bias tends to be negative (i.e., estimates tend to underestimate the parameter), mode A is preferred, since it yields a higher estimate. If the bias tends to be positive, mode B is preferred.

distributions, presence of social desirability responses, length of responses to open-ended questions, length of collection period, and cost of the collection.

De Leeuw and van der Zouwen (1988) have conducted a meta-analysis of 28 research studies comparing face-to-face and telephone interviewing and summarizing the results as if one large study had been conducted. The largest effect was obtained for length of responses to open-ended questions and then in favor of face-to-face interviewing. There also seems to be a tendency for less social desirability bias over the telephone. The large mode effects occur for sensitive variables. For general topics and if no visual communication is needed, telephone interviewing produced about the same quality as face-to-face interviewing.

Any comparison study of two mode systems, A and B, ultimately produces two estimates: one as a result of mode system A and another as a result of mode system B. Even if we have a set of evaluation criteria, it is difficult to assess which mode produces better data quality. Is it the one that produces the smallest measurement error, highest response rate, or that does well on a number of other evaluation criteria? The answer depends on what the researcher believes to be important. Quality components are often in conflict with each other, and even though we are very concerned about data quality, we cannot ignore such factors as cost and the time it takes to produce survey results (see Figure 6.1).

Table 6.3 Mode Effects in a Women's Health Study

	Mean Reported Sex Partners			
Experimental Group	Past Year	Past 5 Years	Lifetime	Percent Admitting Illicit Drug Use
Self-administered questions (SA)	1.72	3.88	6.54	40.9
Conventional SA	1.56	3.37	6.88	42.5
Computer-assisted SA	1.89	4.40	6.25	39.3
Interviewer-administered questions (IA)	1.44	2.82	5.43	40.7
Conventional IA	1.56	2.86	4.58	39.3
Computer-assisted IA	1.36	2.79	6.27	42.2

Source: Adapted from Tourangeau and Smith (1998) and Tourangeau et al. 1997.

Table 6.4 Examples of Estimates of Sensitive Attributes by Mode

A. Percent of U.S. Women Who Have Had One or More Abortions

	Subsequent Reports in ACASI		
Original Report in Interviewer-Administered Questionnaire	No Abortions	1 Abortion	2+ Abortions
No Abortions	95.4	3.5	1.0
1 Abortion	1.8	92.5	5.8
2 Abortions	0.4	2.2	97.4

B. Percentages of U.S. Males Ages 15–19 Reporting Male–Male Sexual Contacts

Mode	Reporting Male–Male Sexual Contacts (%)
Self-administered PAPI	1.1
ACASI	4.7

C. Estimates of Prevalence of Sensitive Behaviors in the U.S. Population

Mode	Having Limited Sexual Experience (No Sex in Past 6 Months) (%)	Most Recent Sexual Relationship Lasted Less Than 6 Months (%)
CAPI	1.5	5.8
T-ACASI	8.0	21.3

Source: Adapted from Turner et al. (1996, 1998).

Perhaps the main problem with mode studies is that we seldom know much about the measurement bias. One way to get information is to compare estimates A and B with a gold standard, as discussed in Section 6.1.6. Such an evaluation study is quite possible to conduct, but the result might be unreliable since evaluation studies are expensive and therefore can be conducted on a limited scale only. An alternative route is to assume a likely direction of the measurement bias. In general, sensitive items are underreported and there is overwhelming evidence that modes that eliminate social desirability bias produce more reporting.

For an illustration of this phenomenon, see Tables 6.3 and 6.4. Here it is assumed that more reporting is better reporting, and for sensitive variables this seems to be a general truth. But no rule is without exceptions. The perception of what might constitute a sensitive variable might very well vary between age and gender groups. For instance, there is evidence that men in their teens, when asked about sexual experiences, tend to overreport. There are other variables considered sensitive where the reporting pattern might vary between groups. Income is such a variable, where the social desirability bias is positive, and as a result, members of some income groups might overreport their income. Thus, to be sure about measurement biases, one has to estimate them by means of evaluation studies.

Examples of other studies include Cannell et al. (1987), who found that question order effects can vary between telephone and face-to-face interviewing. Also, reports of health events were consistently higher in telephone interviews than in face-to-face interviews. Substantive differences between the modes were, however, small. Sykes and Collins (1988) conducted three experimental studies comparing face-to-face and telephone interviewing in Britain. Among other things, they found that answers to open-ended questions tended to be shorter in the telephone mode, which could partially explain the faster pace in the telephone mode. The issue of mode is both complicated and not so complicated. It is complicated in the sense that a pure mode effect is very difficult to assess. It is not so complicated in the sense that in practice the choice of mode is often straightforward, due to design constraints such as cost or topic. Often, modes have to be combined to accomplish acceptable response rates and cost-efficiency, but for some topics and questions, some modes are simply not suitable.